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# Long-tailed classification of pathologies in chest X-rays

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***Abstract*—Rare pathologies are often underrepresented in chest X-ray datasets, making them an important challenge for automated classification. This study investigates whether long-tailed learning methods improve their recognition. Six training strategies are compared on NIH ChestX-ray14 using the same ResNet-152 architecture, patient-disjoint data splits, and three random seeds. Mean average precision (mAP) is the primary metric. Balanced accuracy (bACC), calculated at thresholds selected on validation data, provides a secondary measure. Binary cross-entropy (BCE) achieves the highest overall mAP. Balanced MixUp has the highest mean tail mAP, but the gain comes entirely from hernia. Examining individual errors reveals further differences between the methods. Balanced MixUp detects more labelled pneumonia cases while producing more false alarms. Classifier retraining (cRT) achieves the highest tuned bACC by reducing false positives, although it misses more positive cases. Across methods, pneumonia predictions increase when edema is recorded, including in images without a pneumonia label. These findings show that similar average scores can conceal different patterns of error. Evaluating individual pathologies and their co-occurring labels is therefore necessary to understand whether a long-tailed method improves rare-pathology recognition.**

***Index Terms*—chest X-ray classification, long-tailed learning, multi-label classification, rare pathologies, average precision**

## I. INTRODUCTION

Deep learning models can predict several thoracic findings from a single chest radiograph [Wang et al. 2017; Rajpurkar et al. 2017]. For these models to be useful, they must recognise uncommon conditions as well as frequent findings. A high average score gives an incomplete picture if the same model repeatedly misses a rare pathology.

In a long-tailed dataset, a few classes account for many training examples, while the remaining classes are represented by substantially fewer examples [Zhang et al. 2023]. In this report, head, medium, and tail refer to groups of pathologies with high, intermediate, and low training frequencies. Chest X-ray classification is also *multi-label*: an image can have several pathology labels at the same time. Sampling an image more often to increase exposure to a rare label also increases exposure to the other labels in that image. In addition, most label entries in an image are negative. Class imbalance and label co-occurrence therefore complicate the application of methods developed for single-label recognition.

Ideally, balancing the training process would improve recognition of rare pathologies while preserving performance on common findings. However, balancing changes the examples shown to the model, the contributions to its loss, or the way its classifier is trained. These changes can introduce trade-offs. Previous chest X-ray benchmarks provide a reason to examine those effects directly [Holste et al. 2022 2024]. In particular, an average score does not show whether a method detects more positive cases, produces fewer false alarms, or helps only some pathologies. This study examines these differences alongside the overall performance of each method.

The aim is to identify effective strategies for rare-pathology recognition on NIH ChestX-ray14. Three questions guide the analysis:

1) How does baseline performance differ across head, medium, and tail pathologies, and does frequency alone explain the differences?
2) Which strategies improve tail ranking, and what trade-offs do they introduce between missed cases and false alarms?
3) How do co-occurring labels and extreme prediction scores help characterise errors that aggregate metrics conceal?

The study has four objectives: establish a binary cross-entropy (BCE) baseline across the frequency groups; compare the selected long-tailed training strategies under a shared experimental setup; evaluate both prediction ranking and threshold-based decisions; and examine errors for individual pathologies and co-occurring labels. Mean average precision (mAP) measures ranking quality, while balanced accuracy (bACC) summarises decisions made at thresholds selected on validation data.

The main contributions are the comparison of six training strategies on the same data splits and backbone, and an error analysis that explains differences hidden by their average scores. The findings show that BCE has the highest overall mAP, while Balanced MixUp has an uncertain mean tail gain concentrated in hernia. Classifier retraining (cRT) leads in tuned bACC, but this advantage accompanies more missed positive cases. Together, these results show why a method's effect on rare findings needs to be assessed at the pathology level.

Section II reviews the relevant literature. Section III describes

the data, methods, and training procedure. Section IV presents the evaluation and error analysis, followed by interpretation and limitations in Section V. Section VI concludes the report.

## II. Related Work

### A. Chest X-ray classification and evaluation

Convolutional neural networks learn features from images, and residual shortcut connections make deep networks easier to optimise [He et al. 2016]. Separate sigmoid outputs allow a model to predict several pathologies in the same image. A softmax instead distributes probability across competing classes and is commonly used for single-label classification. Transfer learning supplies an initial set of image features when examples for the target task are limited.

NIH ChestX-ray14 grew from the hospital-scale database of Wang et al. [2017]. Its labels were extracted from radiology reports, making large-scale training possible. Studies of the dataset have identified disagreement between labels and visible findings [Oakden-Rayner 2020]. Broad labels can also hide subgroups of cases that a model recognises poorly [Oakden-Rayner et al. 2020]. These limitations support examining errors for individual pathologies alongside overall performance.

Precision–recall analysis is useful for imbalanced data because it shows how false positives change as more positive cases are retrieved [Saito and Rehmsmeier 2015]. Ranking and binary decisions provide related but different information: a model may rank cases similarly to another model while producing different decisions at the selected thresholds. This study therefore evaluates both ranking quality and decisions made at validation-selected thresholds.

### B. Strategies for long-tailed learning

Long-tailed methods commonly change the loss function or the frequency with which training examples are sampled. Reweighting increases the contribution of classes with fewer examples to the loss [Cui et al. 2019], whereas resampling presents their images more often. In multi-label data, these changes must account for labels that occur together. Increasing exposure to one pathology also changes exposure to the other findings in the same images.

Focal loss reduces the contribution of examples that the model already classifies confidently and gives more attention to harder examples [Lin et al. 2017]. This is relevant to chest X-rays because most label entries are negative, and many are easy to classify. However, a hard example does not necessarily belong to a rare class. Focal loss therefore tests whether reducing the influence of easy examples also benefits less frequent findings.

Distribution-balanced loss (DB loss) addresses label co-occurrence and the large number of negative labels together [Wu et al. 2020]. Its weights account for positive labels that occur in the same image. It also adjusts the contribution of negative labels so that they do not suppress prediction scores as strongly. These features make DB loss relevant to long-tailed, multi-label chest X-ray classification.

Resampling changes which examples the model sees during training. Partial balancing provides a compromise between sampling from the original image distribution and giving each class equal exposure. Because an image retains all its labels when sampled, the effect extends beyond the particular label used to select it. Section III describes the sampling rules used in this study.

Classifier retraining separates feature learning from balancing. Kang et al. [2020] improved long-tailed recognition on ImageNet-LT, Places-LT, and iNaturalist by adjusting classifiers while retaining features learned from the natural training distribution. Here, cRT tests whether changing the classifier is sufficient to improve recognition using the existing chest X-ray features. Its second stage keeps those features fixed, so the adjustment takes place in the mapping from features to predictions.

MixUp generates additional training examples by combining two images and their corresponding labels, with the aim of improving model generalisation [Zhang et al. 2018]. Balanced MixUp combines an ordinary sampling stream with a balanced stream, increasing exposure to rare classes while retaining examples from the original distribution. It therefore changes both the examples presented to the model and the image–target pairs used for learning. Previous applications to retinal and gastrointestinal imaging provide a basis for investigating Balanced MixUp in other medical image classification tasks [Galdran et al. 2021].

### C. Positioning this study

Holste et al. [2022] introduced single-label long-tailed thorax benchmarks derived from NIH and MIMIC-CXR. cRT achieved the highest bACC on both. The later CXR-LT challenge studied 26-label prediction and used macro-averaged average precision as its primary metric [Holste et al. 2024]. CXR-LT 2024 expanded coverage to 45 labels and included manually annotated and zero-shot evaluation tasks [Lin et al. 2025]. These studies inform the choice of methods and metrics, while their different label sets and tasks need to be considered when comparing findings.

The present study retains the original 14 labels and uses a shared ResNet-152 to examine the effects of the training strategies. It builds on these benchmarks by comparing performance across frequency groups and then examining individual pathologies, missed cases, false alarms, and co-occurring labels. This links the method comparison to the practical question of which findings improve and what trade-offs accompany those changes.

## III. Methodology

### A. Dataset and patient-disjoint splits

NIH ChestX-ray14 [Wang et al. 2017] contains 112,120 frontal radiographs from 30,805 patients, with 14 pathology labels. The dataset is suitable for this study because it contains both large differences in class frequency and images with several findings. Each image has a 14-dimensional binary target vector indicating which pathologies are recorded. “No Finding” is represented by an all-zero vector.

The official test split contains 25,596 images from 2,797 patients. The remaining data were divided into 75,505 training

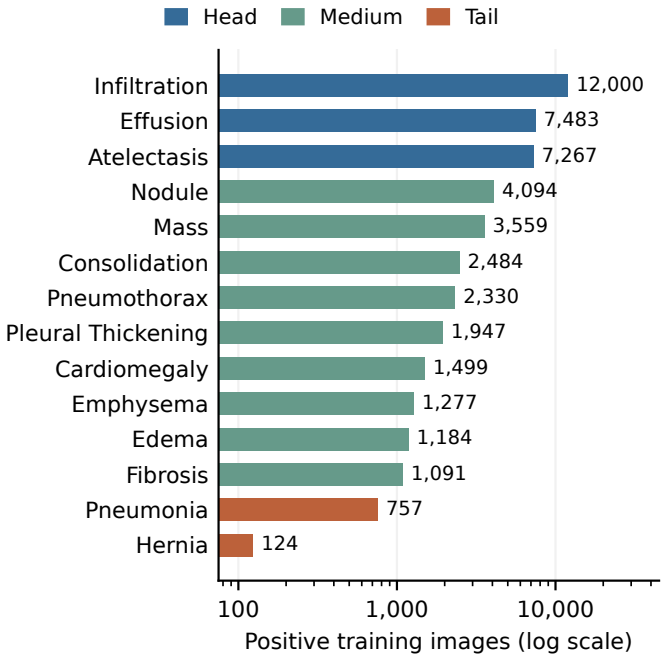



Fig. 1. Positive training images for each pathology. Colours indicate head, medium, and tail groups. An image can contribute to several bars. The logarithmic axis keeps the least frequent labels visible.

images from 24,507 patients and 11,019 validation images from 3,501 patients. To choose the validation set, 200 random patient partitions were compared. The selected partition had the closest match to the patient-level label prevalence of the remaining dataset. The dataset was partitioned at the patient level, ensuring that each patient's images belong to only one split. These partitions were kept consistent across all methods.

Pathologies are grouped by their training frequencies. Head classes have more than 5,000 positive images, medium classes have 1,000–5,000, and tail classes have fewer than 1,000. Atelectasis, effusion, and infiltration form the head group; hernia and pneumonia form the tail group; and the other nine classes form the medium group. Figure 1 shows the distribution. The training set also contains 44,047 all-negative images (58.34%), meaning that none of the pathology labels is recorded. Images with several findings keep their complete label vectors throughout training.

### B. Preprocessing and shared architecture

The original $1024 \times 1024$ images are converted to greyscale, copied across three channels, and resized to $256 \times 256$ pixels using bilinear interpolation. Training uses a random $224 \times 224$ crop, horizontal flipping with probability 0.5, rotation within $\pm 10°$, and brightness and contrast jitter of 0.2. Validation and testing use a centre crop without random augmentation. The channels are normalised using ImageNet means $(0.485, 0.456, 0.406)$ and standard deviations $(0.229, 0.224, 0.225)$. Balanced MixUp applies these same transformations before blending image pairs.

All methods use ResNet-152 with torchvision's ImageNet-1K V2 weights. The original classifier is replaced by a linear layer that maps 2,048 pooled features to 14 logits. A sigmoid function converts each logit into a separate pathology score. Using the same backbone architecture [He et al. 2016] across methods allows the comparison to focus on training strategies.

### C. Loss-based approaches

*1) Binary cross-entropy:* Let $y_{ic} \in \{0, 1\}$ denote the label for image $i$ and class $c$, and let $p_{ic} = \sigma(z_{ic})$ be its predicted score. Here, $z_{ic}$ is the model's logit and $\sigma$ is the sigmoid function. The baseline minimises unweighted binary cross-entropy (BCE):

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{NC} \sum_{i=1}^{N} \sum_{c=1}^{C} [y_{ic} \log p_{ic} + (1 - y_{ic}) \log(1 - p_{ic})], \tag{1}$$

where $N$ is the number of training images and $C = 14$. BCE gives equal weight to each image–label entry. The baseline uses the natural training distribution, including images with no recorded pathology, and provides the reference for assessing the long-tailed methods.

*2) Focal loss:* Focal loss reduces the contribution of image–label entries that the model already classifies correctly with a high score for the observed label. Define $p_{ic}^{t} = y_{ic} p_{ic} + (1 - y_{ic})(1 - p_{ic})$. The loss becomes

$$\mathcal{L}_{\mathrm{FL}} = -\frac{1}{NC} \sum_{i,c} (1 - p_{ic}^{t})^{\gamma} \log p_{ic}^{t}. \tag{2}$$

Larger $\gamma$ places less weight on easy examples. The additional positive/negative $\alpha$ weighting is disabled. Validation selects $\gamma = 2$ from $\{1, 2, 5\}$, following the focusing approach of Lin et al. [2017].

*3) DB loss:* DB loss adjusts the contribution of each image–label entry using class frequency and the labels that occur together. If $n_c$ is the number of positive training images for class $c$, the ratio and its mapped weight are

$$r_{ic} = \frac{1/n_c}{\max(\sum_j y_{ij}/n_j, \epsilon)}, \qquad w_{ic} = 0.1 + \sigma\{10(r_{ic} - \beta)\}. \tag{3}$$

The denominator accounts for all positive labels in an image. We set $\epsilon = 10^{-6}$ to prevent division by zero. The parameter $\beta$ controls the mapping from the ratio to the weight. With $\pi_c = n_c/N$, the implementation shifts logits by $b_c = 0.05 \log[\pi_c/(1 - \pi_c)]$. Writing $u_{ic} = z_{ic} + b_c$, the loss is

$$\mathcal{L}_{\mathrm{DB}} = \frac{1}{NC} \sum_{i,c} w_{ic} \left[ y_{ic} \log(1 + e^{-u_{ic}}) + \frac{1 - y_{ic}}{s} \log(1 + e^{s u_{ic}}) \right]. \tag{4}$$

Here $s$ scales the negative logits, and the reciprocal factor adjusts their contribution to the loss. The selected values are $s = 5$ and $\beta = 0.3$. The weighting and logit adjustments follow the DB loss formulation of Wu et al. [2020].

DB loss uses natural sampling in this study instead of the class-aware sampling used by Wu et al. [2020]. All training images are retained, including those with no recorded pathology.

### D. Resampling, classifier retraining, and mixing

*1) Power-law resampling:* Power-law resampling changes how often images are presented during training while retaining BCE as the loss. For an image with positive-label set $Y_i$, the sampling weight is

$$a_i = \begin{cases} (\min_{c \in Y_i} n_c)^{-q}, & Y_i \neq \varnothing, \\ (\text{median}_c\, n_c)^{-q}, & Y_i = \varnothing. \end{cases} \tag{5}$$

Weights are normalised to mean one, and $N$ images are drawn with replacement per epoch. For an image with several findings, the least frequent positive label determines its weight. All-negative images receive a weight based on the median class count, so they remain available for sampling. Every selected image keeps all its labels.

Square-root sampling corresponds to $q = 0.5$. Validation selects $q = 0.75$ from $\{0.25, 0.5, 0.75\}$, so the results are labelled *power sampling*. This distinction makes clear which sampling strength was evaluated.

*2) Classifier retraining:* Classifier retraining separates feature learning from classifier adjustment. Stage one uses the BCE model trained with the same seed. In stage two, the final linear layer is reinitialised, while the backbone and its batch-normalisation statistics are frozen. Only the classifier is trained, using BCE.

Each image with a positive label is assigned to a bucket for its rarest positive pathology. All-negative images form an additional bucket. Sampling first chooses a non-empty bucket uniformly, then samples an image uniformly with replacement from that bucket, for $N$ draws per epoch. The full label vector is retained even though one label determines the bucket. This adapts cRT [Kang et al. 2020] to multi-label targets and tests whether balancing the classifier helps when the image features remain unchanged.

*3) Balanced MixUp:* Balanced MixUp combines a batch sampled from the natural distribution with a partner batch from the bucket-balanced scheme used by cRT. Both images and label vectors are blended. For ordinary example $(x_o, y_o)$ and balanced example $(x_b, y_b)$,

$$\widetilde{x} = (1-\lambda)x_o + \lambda x_b, \quad \widetilde{y} = (1-\lambda)y_o + \lambda y_b, \quad \lambda \sim \text{Beta}(\alpha, 1). \tag{6}$$

One mixing coefficient is sampled per batch. The resulting target entries can lie between absent and present, and BCE is applied to these soft targets. The selected $\alpha = 0.3$ gives an expected balanced-stream weight of $\mathbb{E}[\lambda] = 0.3/1.3 \approx 0.231$. Ordinary images therefore make the larger contribution on average, while the balanced stream increases exposure to rare findings. This adaptation follows the rationale of Galdran et al. [2021], combining balanced sampling with image mixing.

### E. Training and model selection

Training uses PyTorch and torchvision with CUDA on a computing cluster. Each job requests four CPU cores and 48 GB RAM.

All methods use AdamW [Loshchilov and Hutter 2019] with batch size 16, weight decay $10^{-4}$, gradient clipping at norm 1.0, and mixed precision. The optimiser uses its default momentum parameters $(0.9, 0.999)$ and $\epsilon = 10^{-8}$. Every selected configuration has an initial learning rate of $10^{-4}$.

TABLE I
VALIDATION-ONLY SEARCH AND SELECTED SETTINGS.

| Method | Candidate values | Selected |
|---|---|---|
| BCE learning rate | $3\times10^{-5}, 10^{-4}, 3\times10^{-4}$ | $10^{-4}$ |
| Focal $\gamma$ | $1, 2, 5$ | 2 |
| DB loss $(s, \beta)$ | $\{1, 2, 5\} \times \{0.2, 0.3\}$ | $(5, 0.3)$ |
| Sampling $q$ | $0.25, 0.5, 0.75$ | 0.75 |
| cRT learning rate | $10^{-4}, 3\times10^{-4}, 10^{-3}$ | $10^{-4}$ |
| MixUp $\alpha$ | $0.1, 0.2, 0.3$ | 0.3 |

The learning rate is halved after three epochs without an improvement in validation mAP. Training is limited to 30 epochs. Early stopping uses a patience of seven epochs, and the checkpoint with the highest validation mAP is retained.

The hyperparameters included in the sweeps were motivated by work on focal loss, DB loss, classifier balancing, and Balanced MixUp [Lin et al. 2017; Wu et al. 2020; Kang et al. 2020; Galdran et al. 2021]. Table I lists the candidate values and selected settings. Candidates are compared using seed 42, with the other settings held fixed within each method. The configuration with the highest validation mAP is selected; ties are resolved by choosing the first candidate.

Each selected configuration is then trained with seeds 42, 123, and 2024, giving 18 final runs. Training starts again from the pretrained weights, except that cRT uses the BCE backbone trained with the corresponding seed. The repeated runs show how much the results vary with training randomness.

The test set is reserved for final evaluation. Once training is complete, the decision thresholds are optimised for bACC using validation data only, following the procedure described in Section IV. This keeps model selection and threshold selection separate from the reported test performance.

## IV. Experiments and Results

### A. Evaluation metrics and uncertainty

For each pathology, average precision (AP) summarises precision as recall increases across distinct score thresholds:

$$\text{AP}_c = \sum_k (R_{c,k} - R_{c,k-1}) P_{c,k}, \qquad \text{mAP}_G = \frac{1}{|G|} \sum_{c \in G} \text{AP}_c. \tag{7}$$

Here, $P$ is precision and $R$ is recall. Images with the same prediction score are processed together, using non-interpolated AP. Overall mAP is the mean AP across all 14 classes. Group mAP uses the same calculation within the head, medium, or tail group. Results are presented as percentages, so a percentage-point change is an absolute difference on the 0–100 scale.

mAP is the primary metric because it evaluates the ranking of images without requiring a fixed decision threshold. For each pathology, AP summarises the precision achieved as more positive cases are retrieved. Taking the mean across pathologies gives each class equal weight, even when its training examples are rare. This choice follows the CXR-LT benchmark [Holste et al. 2024] and the use of precision–recall analysis for imbalanced data [Saito and Rehmsmeier 2015].

Balanced accuracy (bACC) is the secondary metric. For each pathology $c$, it gives equal weight to recognising positive and negative cases:

$$\mathrm{bACC}_c(t) = \frac{1}{2}\left[\frac{\mathrm{TP}_c(t)}{\mathrm{TP}_c(t)+\mathrm{FN}_c(t)} + \frac{\mathrm{TN}_c(t)}{\mathrm{TN}_c(t)+\mathrm{FP}_c(t)}\right]. \tag{8}$$

TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives. The first fraction is sensitivity: the proportion of positive cases detected. The second is specificity: the proportion of negative cases correctly rejected. A score greater than or equal to $t$ produces a positive prediction. Overall bACC is the unweighted mean across the 14 pathologies. Every pathology has both positive and negative cases in the validation and test sets.

bACC is reported at validation-tuned thresholds. For each method, seed, and pathology, the tuned threshold is

$$t_c^* = \arg\max_t \ \mathrm{bACC}_c^{\mathrm{val}}(t). \tag{9}$$

The search considers every distinct set of decisions on the validation data, including the decision to predict all images as negative. Equal scores are grouped together. If several thresholds give the same maximum bACC, the highest threshold is selected. Each selected threshold is then applied unchanged to the corresponding test predictions. Test labels are not used to choose thresholds.

Tables report means and sample standard deviations across three training seeds. For AP and mAP, uncertainty is estimated using 1,000 patient-bootstrap resamples with random seed 42. Each resample draws patients with replacement and retains all images belonging to each selected patient. This keeps images from the same patient together. The same resamples are used for every method and seed, allowing paired comparisons with BCE.

Within each resample, the AP or mAP statistic is calculated for each of the three seed models and then averaged. A paired difference is this average minus the corresponding BCE average. The 2.5th and 97.5th percentiles form the 95% intervals. These intervals describe variation associated with the test patients, while the standard deviations across seeds describe variation associated with training. bACC is reported using its mean and standard deviation across seeds.

*1) Error analysis:* A false negative occurs when a model misses a recorded positive pathology label. A false positive occurs when it predicts a pathology whose label is negative. The false negative rate (FNR) and false positive rate (FPR) are $\mathrm{FNR}_c = \mathrm{FN}_c/(\mathrm{TP}_c+\mathrm{FN}_c)$ and $\mathrm{FPR}_c = \mathrm{FP}_c/(\mathrm{TN}_c+\mathrm{FP}_c)$. These errors are measured against the dataset labels and use the same validation-tuned thresholds as bACC.

Rates are calculated separately for each pathology and seed, then averaged within frequency groups and across seeds. Each pathology therefore receives equal weight. A common label does not dominate the group result simply because it has more examples.

The co-occurrence analysis compares error rates when another recorded label is present and when it is absent. Both subgroups must contain at least 50 relevant positive or negative images and 20 distinct patients. Selected differences are checked with 1,000 paired patient-bootstrap resamples, with the models and thresholds held fixed.

The analysis also examines errors with extreme prediction scores: positive labels scoring at most 0.1 and negative labels scoring at least 0.9. A case counts as an error only if its score also lies on the wrong side of its tuned threshold.

### B. Frequency distribution and overall results

The training distribution in Figure 1 has an imbalance ratio of $12{,}000/124 = 96.77$. The two tail labels account for only 1.87% of positive training-label assignments. This distribution provides the context for the test results in Table II.

BCE has the highest overall and head mAP, with a head–tail gap of 18.72 percentage points. Balanced MixUp has the highest mean tail mAP, while focal loss remains close to BCE in each frequency group. The selected DB loss and power-sampling configurations have lower overall mAP than BCE in all three seeds. cRT leads in validation-tuned bACC, the secondary metric.

Balanced MixUp and BCE are closely matched overall. The paired difference in mAP is $-0.04$ percentage points, with a 95% patient-bootstrap interval of $[-0.41, 0.35]$. Focal loss, DB loss, and cRT also achieve similar overall mAP to BCE, while power sampling shows a larger reduction in performance.

Figure 2 presents the paired changes for the head and tail groups. Balanced MixUp's tail change is $+1.54$ points $[-0.56, 3.55]$, and its head change is $-0.19$ points $[-0.54, 0.15]$. Both intervals include zero. The estimated tail gain is therefore uncertain, and the pathology-level results are needed to show where it occurs.

### C. Pathology-level results and operating thresholds

The pathology-level results in Table III show that Balanced MixUp's tail gain is concentrated in hernia. Hernia AP increases by 3.19 percentage points, while pneumonia AP remains slightly below BCE. The mean tail improvement therefore comes entirely from hernia, and the tail difference is not positive in every seed. Balanced MixUp also improves the mean AP of fibrosis and pneumothorax, which belong to the medium group.

Hernia has 86 positive test images (0.34%), compared with 555 (2.17%) for pneumonia. Every method achieves higher AP for hernia, despite its smaller training set. Fibrosis also has low AP, while emphysema and pneumothorax have higher AP than the more frequent infiltration label. Relative to BCE, cRT reduces infiltration AP by 4.34 points. DB loss has higher mean AP than BCE for edema and hernia, but lower AP for the other pathologies.

cRT's leading tuned bACC is 0.49 percentage points above BCE. Because bACC combines sensitivity and specificity, this difference could reflect changes in either the detection of positive cases or the rejection of negative cases. The error analysis separates these outcomes.

### D. Error analysis

*1) Missed cases across frequency groups:* At the tuned thresholds, every method has a higher mean FNR for tail than

TABLE II
TEST PERFORMANCE (%, MEAN ± SAMPLE STANDARD DEVIATION OVER THREE SEEDS). MAP IS PRIMARY; BACC USES PATHOLOGY THRESHOLDS SELECTED ON VALIDATION DATA. BOLD MARKS THE HIGHEST MEAN IN EACH COLUMN. POWER SAMPLING USES $q = 0.75$.

| Method | Overall mAP | Head mAP | Medium mAP | Tail mAP | bACC tuned |
|---|---|---|---|---|---|
| BCE | **28.37 ± 0.34** | **40.91 ± 0.11** | 25.55 ± 0.37 | 22.19 ± 1.41 | 72.56 ± 0.21 |
| Focal loss | 28.20 ± 0.30 | 40.58 ± 0.15 | 25.39 ± 0.34 | 22.27 ± 2.34 | 72.70 ± 0.41 |
| DB loss | 28.11 ± 0.35 | 40.61 ± 0.32 | 25.17 ± 0.27 | 22.56 ± 1.64 | 72.03 ± 0.20 |
| Power sampling | 26.55 ± 0.52 | 39.32 ± 0.50 | 23.44 ± 0.22 | 21.40 ± 3.48 | 71.43 ± 0.21 |
| cRT | 28.18 ± 0.39 | 39.74 ± 0.78 | **25.66 ± 0.11** | 22.17 ± 1.22 | **73.05 ± 0.36** |
| Balanced MixUp | 28.33 ± 0.29 | 40.73 ± 0.05 | 25.21 ± 0.13 | **23.74 ± 1.48** | 72.68 ± 0.19 |

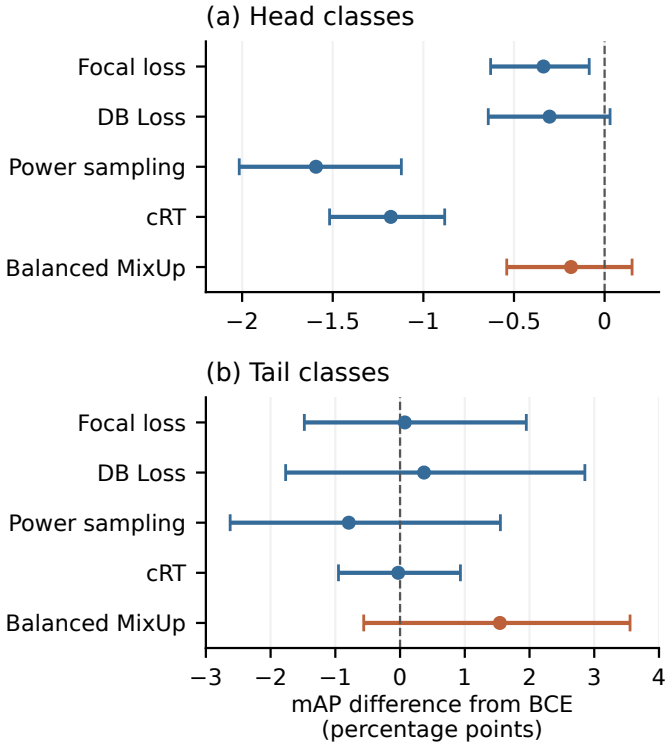



Fig. 2. Changes from BCE in head and tail mAP. Points are three-seed mean differences; bars are 95% patient-bootstrap intervals. Positive values favour the indicated method.

head pathologies. For BCE, the mean miss rate is 24% for the tail, 17% for the head, and 18% for the medium group. The BCE tail–head difference remains positive in the patient-bootstrap analysis. Head classes have higher mean false positive rates: they are detected more often, but also produce more false alarms.

The two tail labels differ from each other. Pneumonia is missed more often than hernia, even though it has more training examples. Nodule and fibrosis also have high miss rates, showing that errors are not confined to the tail group.

*2) Changes in missed cases and false alarms:* Across all 14 pathologies, cRT has a higher mean FNR and a lower mean FPR than BCE. Its higher bACC therefore accompanies fewer false alarms and more missed positive cases. Nodule shows a particularly large change. cRT misses about half of the labelled positive images, compared with 27% for BCE, while reducing FPR from 35% to 16%. Both directions hold in every seed and in the paired patient-bootstrap checks. DB loss has a nodule miss rate of about 27%, close to BCE, and a mean tail miss rate of 23%.

For pneumonia, Balanced MixUp reduces the mean FNR by 4.4 percentage points relative to BCE and increases FPR by 5.7 points. Both directions hold in all three seeds. It detects more labelled pneumonia cases at the selected thresholds, although its pneumonia AP remains slightly below BCE. Power sampling has a similar average trade-off, but its pneumonia differences change direction across seeds.

*3) Errors when other labels are present:* Pneumonia is detected more often when edema is also recorded. For BCE, the pneumonia miss rate falls from 41% without edema to 15% with it. Among pneumonia-negative images, however, FPR rises from 34% to 72% when edema is present. All six methods show both directions in every seed. The presence of an edema label is therefore associated with more positive pneumonia predictions, including more predictions that disagree with the recorded pneumonia label.

Effusion and infiltration show a different pattern depending on which label is being predicted. Across methods, effusion is missed slightly less often when infiltration is recorded, while effusion false alarms become more common. In the reverse comparison, BCE misses infiltration less often when effusion is present, whereas cRT misses it more often. This reversal holds across seeds and in the patient-bootstrap checks.

*4) Errors with extreme prediction scores:* cRT has a lower mean emphysema FPR than BCE, but more false positives with very high scores. Across seeds, an average of 170 emphysema-negative images receive cRT scores of at least 0.9, compared with 43 for BCE. The increase occurs in every seed. A lower overall false alarm rate therefore does not imply fewer errors at extreme scores.

The selected threshold also matters when interpreting low scores on positive images. A score below 0.1 can still give a correct positive prediction if the pathology's threshold is lower than that score.

## V. DISCUSSION

### A. What the tail average reveals

The first research question asks how performance differs across frequency groups and whether rarity explains those differences. The results show a clear practical concern: every method has a higher mean miss rate for tail than head pathologies at the selected thresholds. Even the leading overall

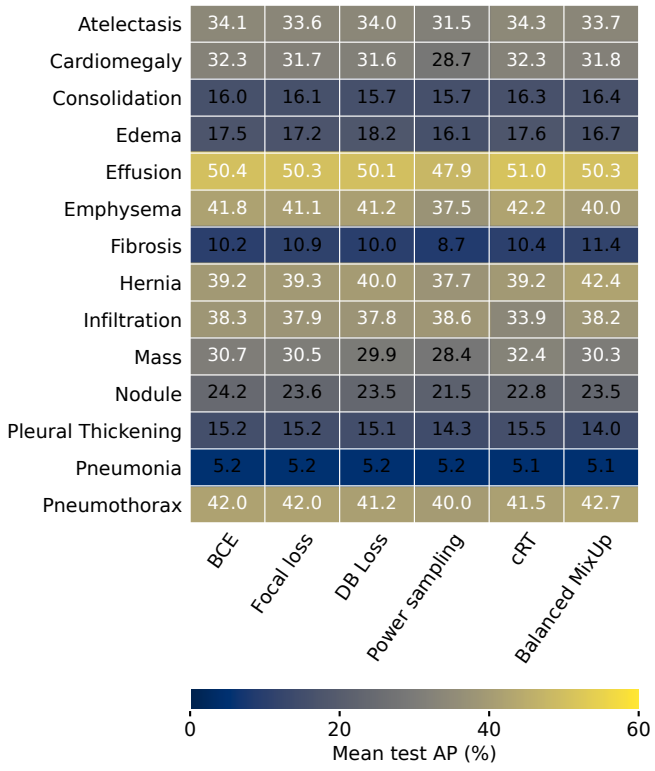



Fig. 3. Per-class test AP (%, mean over three seeds).

method, BCE, has this pattern. A strong average score therefore gives an incomplete account of rare-pathology recognition.

The heat map in Figure 3 shows why frequency groups need to be examined alongside individual pathologies. Hernia has fewer training examples than pneumonia, yet every method achieves higher AP for hernia. Fibrosis and nodule also remain difficult despite belonging to the medium group. Training frequency is therefore insufficient to describe the observed differences in performance. This comparison does not isolate the effect of rarity, because the pathologies differ in other ways and AP also depends on their test prevalence.

The pneumonia–edema results provide further context for the third research question. When edema is recorded, the models produce more positive pneumonia predictions on both pneumonia-positive and pneumonia-negative images. Previous work has documented label disagreement and hidden subgroups in chest X-ray datasets [Oakden-Rayner 2020; Oakden-Rayner et al. 2020]. These findings make such explanations plausible, but the present analysis cannot distinguish between them.

With only two tail labels, improvement on one pathology can raise tail mAP while leaving the other unresolved. Balanced MixUp illustrates this directly: its mean tail gain comes from hernia. The interval for the tail-mAP difference includes no improvement, and the gain is not positive in every seed. The evidence therefore supports a possible benefit for hernia, rather than a general conclusion that Balanced MixUp solves rare-pathology recognition.

### *B. Balancing changes ranking and decisions*

The second research question concerns the benefits and trade-offs of long-tailed training. Balanced MixUp shows why the answer depends on both the pathology and the metric. It improves the mean ranking of hernia, but its main effect on pneumonia at the selected thresholds is to detect more positive cases while producing more false alarms. Its combination of balanced exposure and image mixing [Galdran et al. 2021] therefore has different outcomes for the two rare findings.

This distinction follows from what the metrics measure. AP considers predictions across score thresholds, whereas FNR and FPR describe decisions at the chosen threshold. Detecting more pneumonia cases at that threshold can coexist with slightly lower pneumonia AP. The results are therefore consistent: they describe different aspects of the same predictions. Assessing both avoids treating a change in operating behaviour as an improvement in ranking.

Across the methods, the results do not establish a consistent advantage over BCE for both rare pathologies while preserving overall ranking. They instead identify particular benefits and costs that an overall average would conceal.

### *C. Why power sampling underperformed*

Power sampling achieved lower overall mAP than BCE across all three seeds. Its sampling distributions help explain these results, as both change the frequency and diversity of images seen during training.

Power sampling increases the proportion of all-negative images from 58.3% to 65.0% and reduces the diversity of positive training examples. Its validation mAP peaks at epochs 2–3 before declining, suggesting possible early overfitting.

### *D. Understanding cRT's higher balanced accuracy*

cRT provides another answer to the second research question. Its higher bACC comes from reducing false alarms enough to offset an increase in missed positive cases. The nodule results make this trade-off particularly clear. A higher bACC therefore does not imply that the model detects more disease-positive images.

Because cRT adjusts the classifier while keeping the image features fixed [Kang et al. 2020], it changes how the existing representation is used to make predictions. The infiltration–effusion reversal shows that these changes can affect particular combinations of labels. The emphysema analysis adds a separate concern: fewer false alarms overall can coexist with more false positives at very high scores.

Taken together, the results support using bACC as a secondary measure alongside mAP. It summarises the balance between sensitivity and specificity, while the error rates show how that balance was achieved. This makes the reported advantage more informative than a ranking of methods by bACC alone.

### *E. Comparison with Holste's benchmark*

Holste et al. [2022] found that cRT achieved the highest bACC on both NIH-CXR-LT and MIMIC-CXR-LT. cRT also

ranks first among the six methods in this study when thresholds maximise validation bACC. Among the three shared methods, NIH-CXR-LT ranks cRT first, followed by Balanced MixUp and focal loss. MIMIC-CXR-LT ranks cRT first, followed by focal loss and Balanced MixUp. The present tuned results follow the latter order, so the strongest agreement is cRT's leading position.

The absolute bACC values cannot be compared directly. Holste's benchmarks use different label sets, single-label images, and a ResNet-50. Their multiclass bACC is the mean class recall. Here, bACC averages binary sensitivity and specificity across 14 labels. Agreement in the leading method is useful context, but it does not make the numerical scores equivalent or demonstrate that cRT improves the same aspect of performance in both settings. The present error analysis identifies the trade-off associated with its leading score on ChestX-ray14.

### *F. Limitations and future work*

The study uses one dataset, one patient partition, and three seeds. Small differences should therefore be interpreted cautiously, and performance on images from other hospitals remains untested [Zech et al. 2018]. The shared architecture and evaluation protocol make comparisons within this study useful, but they do not establish that the same rankings would hold for other datasets or models.

There are also limits to what the data can show. Report-derived labels may disagree with the visible findings, and $224 \times 224$ inputs limit the detail available to the models. Hernia has only 17 positive validation images, which makes its threshold estimate less stable, and its positive co-occurrence subgroups are too small for analysis. AP also depends on class prevalence, so comparisons between pathologies require care.

Hyperparameter tuning was also not extensive, as only a limited range of values was explored. In particular, the learning rate was selected through tuning on the BCE baseline and then used for the other methods. This may have favoured BCE, as other methods could perform better with different learning rates. A broader search may have identified better configurations and improved the performance of some methods.

Future work should begin with targeted image and label review of the pneumonia–edema pattern and the extreme emphysema errors. Evaluating on an external dataset would test whether the findings extend beyond ChestX-ray14.

## VI. Conclusion

This study evaluated different approaches to learning from imbalanced chest X-ray datasets, focusing on how well they identify less frequent pathologies when multiple labels can occur in the same image. BCE achieved the highest overall mAP, Balanced MixUp had the highest mean tail mAP, and cRT led in validation-tuned bACC. However, the tail gain was uncertain and concentrated in hernia. The results do not establish a consistent improvement in rare-pathology recognition across the evaluated methods.

The main contribution is the combination of a shared experimental comparison with an analysis of the errors behind the average scores. cRT reduced false alarms while missing more positive cases, and Balanced MixUp detected more pneumonia cases with additional false positives. Co-occurring labels and extreme prediction scores exposed further differences that the aggregate metrics did not show. Evaluating individual pathologies alongside overall scores is therefore necessary to understand the benefits and limits of balancing.

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## Code Availability

The source code for this study is available at https://github.com/git-gohil/long-tailed-cxr-classification.

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# APPENDIX
# SUPPLEMENTARY PATHOLOGY-WISE RESULTS

## A. Pathology-wise average precision

Table III provides the full numerical results corresponding to the pathology-wise AP heatmap in Figure 3. Results are reported as mean AP (%) $\pm$ sample standard deviation.

## B. Pathology-wise decision thresholds

Table IV reports the decision thresholds selected to maximise validation bACC separately for each method, pathology, and training seed, following the procedure in Section IV. Entries are the mean and sample standard deviation across the three training seeds.

The selected thresholds differ across methods and pathologies. Mean thresholds range from 0.0009–0.1826 for BCE, 0.0654–0.3638 for focal loss, and 0.2195–0.4262 for DB loss. Infiltration has the highest mean threshold within all six methods. These values describe operating cutoffs for each method's prediction scores.

Variation across seeds also differs by pathology. For Balanced MixUp, hernia has the largest sample standard deviation, with thresholds of 0.0082, 0.0480, and 0.0083 across the three seeds, giving a mean of 0.0215 and standard deviation of 0.0229. This variation should be interpreted alongside the limited validation support for hernia, which has 17 positive images. The standard deviations describe variation across three training seeds on the same validation split; they do not quantify uncertainty from choosing a different patient split.

TABLE III

PATHOLOGY AP (%, MEAN ± SAMPLE STANDARD DEVIATION). H/M/T INDICATE TRAINING-FREQUENCY GROUPS. COUNTS ARE POSITIVE TRAINING IMAGES. BOLD MARKS THE HIGHEST MEAN FOR EACH PATHOLOGY.

| Pathology (group) | Train $n$ | BCE | Focal | DB loss | Power sampling | cRT | Balanced MixUp |
|---|---|---|---|---|---|---|---|
| Atelectasis (H) | 7,267 | $34.08 \pm 0.14$ | $33.55 \pm 0.55$ | $33.95 \pm 0.68$ | $31.47 \pm 0.85$ | $\mathbf{34.30 \pm 0.54}$ | $33.69 \pm 0.49$ |
| Cardiomegaly (M) | 1,499 | $\mathbf{32.30 \pm 0.77}$ | $31.74 \pm 0.74$ | $31.55 \pm 0.32$ | $28.66 \pm 0.76$ | $32.26 \pm 0.82$ | $31.83 \pm 1.80$ |
| Consolidation (M) | 2,484 | $15.96 \pm 0.29$ | $16.09 \pm 0.06$ | $15.72 \pm 0.73$ | $15.73 \pm 0.47$ | $16.33 \pm 0.40$ | $\mathbf{16.41 \pm 0.23}$ |
| Edema (M) | 1,184 | $17.49 \pm 1.06$ | $17.22 \pm 1.68$ | $\mathbf{18.18 \pm 0.41}$ | $16.11 \pm 0.53$ | $17.58 \pm 0.54$ | $16.72 \pm 0.89$ |
| Effusion (H) | 7,483 | $50.38 \pm 0.78$ | $50.30 \pm 0.96$ | $50.06 \pm 0.15$ | $47.95 \pm 0.49$ | $\mathbf{50.96 \pm 0.31}$ | $50.29 \pm 0.04$ |
| Emphysema (M) | 1,277 | $41.82 \pm 1.17$ | $41.13 \pm 0.20$ | $41.24 \pm 1.25$ | $37.48 \pm 0.49$ | $\mathbf{42.19 \pm 0.69}$ | $40.04 \pm 0.99$ |
| Fibrosis (M) | 1,091 | $10.24 \pm 0.55$ | $10.94 \pm 0.26$ | $10.05 \pm 0.22$ | $8.71 \pm 0.74$ | $10.39 \pm 0.40$ | $\mathbf{11.38 \pm 0.84}$ |
| Hernia (T) | 124 | $39.21 \pm 2.99$ | $39.31 \pm 4.55$ | $39.97 \pm 3.07$ | $37.65 \pm 6.76$ | $39.21 \pm 2.46$ | $\mathbf{42.40 \pm 2.88}$ |
| Infiltration (H) | 12,000 | $38.29 \pm 0.60$ | $37.89 \pm 0.90$ | $37.82 \pm 0.57$ | $\mathbf{38.56 \pm 0.61}$ | $33.95 \pm 2.29$ | $38.21 \pm 0.39$ |
| Mass (M) | 3,559 | $30.74 \pm 1.53$ | $30.55 \pm 1.62$ | $29.93 \pm 0.95$ | $28.40 \pm 1.47$ | $\mathbf{32.36 \pm 0.63}$ | $30.33 \pm 2.26$ |
| Nodule (M) | 4,094 | $\mathbf{24.23 \pm 0.25}$ | $23.62 \pm 0.19$ | $23.54 \pm 0.16$ | $21.48 \pm 1.87$ | $22.81 \pm 0.51$ | $23.49 \pm 0.35$ |
| Pleural Thickening (M) | 1,947 | $15.24 \pm 0.61$ | $15.21 \pm 0.33$ | $15.10 \pm 0.47$ | $14.33 \pm 0.56$ | $\mathbf{15.54 \pm 0.60}$ | $13.96 \pm 0.58$ |
| Pneumonia (T) | 757 | $5.18 \pm 0.23$ | $\mathbf{5.22 \pm 0.23}$ | $5.15 \pm 0.21$ | $5.15 \pm 0.35$ | $5.12 \pm 0.30$ | $5.08 \pm 0.13$ |
| Pneumothorax (M) | 2,330 | $41.98 \pm 0.09$ | $41.99 \pm 1.30$ | $41.19 \pm 0.59$ | $40.04 \pm 1.86$ | $41.47 \pm 0.98$ | $\mathbf{42.74 \pm 0.67}$ |

TABLE IV

PATHOLOGY-WISE THRESHOLDS FOR VALIDATION-TUNED BACC (MEAN ± SAMPLE STANDARD DEVIATION ACROSS SEEDS 42, 123, AND 2024).

| Pathology | BCE | Focal | DB loss | Power sampling | cRT | Balanced MixUp |
|---|---|---|---|---|---|---|
| Atelectasis | $0.0749 \pm 0.0195$ | $0.2837 \pm 0.0136$ | $0.3846 \pm 0.0058$ | $0.0639 \pm 0.0120$ | $0.1107 \pm 0.0060$ | $0.0983 \pm 0.0179$ |
| Cardiomegaly | $0.0218 \pm 0.0153$ | $0.1826 \pm 0.0636$ | $0.3462 \pm 0.0275$ | $0.0204 \pm 0.0081$ | $0.0673 \pm 0.0198$ | $0.0389 \pm 0.0062$ |
| Consolidation | $0.0297 \pm 0.0094$ | $0.1952 \pm 0.0111$ | $0.3408 \pm 0.0160$ | $0.0394 \pm 0.0081$ | $0.0706 \pm 0.0148$ | $0.0394 \pm 0.0062$ |
| Edema | $0.0195 \pm 0.0126$ | $0.1571 \pm 0.0366$ | $0.3166 \pm 0.0175$ | $0.0247 \pm 0.0055$ | $0.0696 \pm 0.0247$ | $0.0456 \pm 0.0106$ |
| Effusion | $0.0832 \pm 0.0152$ | $0.3062 \pm 0.0341$ | $0.3787 \pm 0.0089$ | $0.0764 \pm 0.0208$ | $0.1677 \pm 0.0078$ | $0.1510 \pm 0.0168$ |
| Emphysema | $0.0117 \pm 0.0039$ | $0.1668 \pm 0.0253$ | $0.3161 \pm 0.0123$ | $0.0303 \pm 0.0046$ | $0.0394 \pm 0.0158$ | $0.0198 \pm 0.0049$ |
| Fibrosis | $0.0091 \pm 0.0020$ | $0.1480 \pm 0.0129$ | $0.3126 \pm 0.0045$ | $0.0247 \pm 0.0074$ | $0.0565 \pm 0.0070$ | $0.0207 \pm 0.0028$ |
| Hernia | $0.0009 \pm 0.0005$ | $0.0654 \pm 0.0189$ | $0.2195 \pm 0.0139$ | $0.0024 \pm 0.0018$ | $0.0566 \pm 0.0154$ | $0.0215 \pm 0.0229$ |
| Infiltration | $0.1826 \pm 0.0327$ | $0.3638 \pm 0.0151$ | $0.4262 \pm 0.0200$ | $0.1332 \pm 0.0303$ | $0.3068 \pm 0.0168$ | $0.2022 \pm 0.0158$ |
| Mass | $0.0476 \pm 0.0111$ | $0.2492 \pm 0.0209$ | $0.3460 \pm 0.0094$ | $0.0542 \pm 0.0130$ | $0.0901 \pm 0.0051$ | $0.0676 \pm 0.0121$ |
| Nodule | $0.0425 \pm 0.0114$ | $0.2275 \pm 0.0097$ | $0.3610 \pm 0.0096$ | $0.0450 \pm 0.0170$ | $0.1147 \pm 0.0307$ | $0.0568 \pm 0.0075$ |
| Pleural Thickening | $0.0198 \pm 0.0029$ | $0.1951 \pm 0.0192$ | $0.3386 \pm 0.0165$ | $0.0258 \pm 0.0057$ | $0.0818 \pm 0.0050$ | $0.0377 \pm 0.0029$ |
| Pneumonia | $0.0142 \pm 0.0024$ | $0.1616 \pm 0.0225$ | $0.3092 \pm 0.0252$ | $0.0225 \pm 0.0130$ | $0.0652 \pm 0.0024$ | $0.0222 \pm 0.0014$ |
| Pneumothorax | $0.0137 \pm 0.0041$ | $0.1847 \pm 0.0059$ | $0.3342 \pm 0.0101$ | $0.0168 \pm 0.0085$ | $0.0636 \pm 0.0189$ | $0.0322 \pm 0.0079$ |